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Thapar Institute of Engineering and Technology (TIET), Patiala
Department of Computer Science & Engineering (CSED)
Mid Semester Examination (MST)

B. E. (Third Year): Semester-VI | **Course Code: UCS677**

March 3, 2025 (3:00 pm) | **Course Name: Data Engineering**

Max. Marks: 30 | **Faculty: Dr. Mandeep Kaur**
Dr. Harpreet Singh

Note: Attempt all the questions in sequence. Assume any missing data. Please be precise and concise while writing.

Q1: A startup company developing a scalable e-commerce platform initially used MySQL to store customer data, product details, and transactions. However, as their user base grew, they faced challenges in handling dynamic product attributes, high-velocity transactions, and horizontal scaling. To address these issues, they are considering migrating to MongoDB, a NoSQL database. Analyze the migration decision and answer the following questions with structured explanations and examples:
Q1(a): Compare MySQL and MongoDB and justify why MongoDB may be better suited for scalability in e-commerce. **(3 Marks)**
Q1(b): How does hierarchical data organization in MongoDB support flexible and scalable data management in dynamic e-commerce applications? **(3 Marks)**
Q1(c): Describe JSON and BSON in MongoDB. Why does MongoDB use BSON instead of plain JSON? **(2 Marks)**

Q2: The recent announcement by Donald Trump regarding the "Gold Card" visa aims to attract skilled professionals, including Indians, by providing a fast-track path to U.S. citizenship and simplifying hiring for U.S. companies.
A U.S.-based technology recruitment company, TechHire Inc., is implementing a MongoDB database to manage job applications, track applicant status, and update candidate records dynamically. To store real-time applicant logs efficiently while retaining only the latest records, they plan to use CRUD operations, capped collections, and update operators. As a database administrator, you have been assigned to perform the following MongoDB tasks and write the query to:
Q2(a): Insert a new applicant into the "Candidates" collection with fields: name, nationality, visa_status, years of experience, and salary expected. **(0.5 Mark)**
Q2(b): Retrieve all applicants from India and delete any applicant whose visa application was rejected. **(1 Mark)**
Q2(c): Create a collection named "ApplicationLogs" with a maximum size of 1MB and a limit of 100 documents to store real-time updates about candidate applications. Give one line answer that why using a capped collection is beneficial for tracking applicant activity. **(1.5 Mark)**
Q2(d): Increase the salary expectation of all candidates by 10%. **(1 Mark)**
Q2(e): Ensure a candidate's years of experience are not reduced below 5. **(1 Mark)**
Q2(f): Change the field name visa_status to immigration_status. **(1 Mark)**
Q2(g): Update a specific candidate's nationality to "U.S." after receiving citizenship. **(1 Mark)**

Q3: How do ACID properties differ from the CAP theorem in database management? Explain the components of the CAP theorem. Describe which CAP properties MongoDB adheres to and the mechanisms it uses to implement them. **(5 Marks)**

Q4: Consider the following "Student" collection:

{ "_id": 1, "name": "James", "courses": [{ "title": "Deep Learning", "progress (in %age)": [10, 30, 60, 90], "completed": false }, { "title": "MongoDB", "progress (in %age)": [100, 100, 100], "completed": true }], "interests": ["AI", "Big Data", "Databases"], "grades": [92, 88, 95, 100], "certifications": ["Python"], "engagement": { "forumPosts": [5, 10, 15], "quizAttempts": [2, 3, 1] } }

{ "_id": 2, "name": "Lee", "courses": [{ "title": "Deep Learning", "progress (in %age)": [20, 50, 90, 100], "completed": true }, { "title": "MongoDB Advanced", "progress (in %age)": [100, 100, 100], "completed": true }], "interests": ["AI", "Data Science", "Cloud"], "grades": [83, 89, 93, 98], "certifications": ["Data Science"], "engagement": { "forumPosts": [5, 8, 13], "quizAttempts": [2, 2, 1] } }

Write queries to answer the following questions:

Q4(a): Find students who have at least one course with progress reaching 90% but have not yet completed the course. **(1.5 Marks)**

Q4(b): Update all students who have completed at least one course by adding "Advanced Learner" to their certifications, but only if they don't already have it. **(1.5 Marks)**

Q4(c): Remove courses from students where all progress values are below 50 and they have never attempted a quiz more than once. **(3 Marks)**

Q5: A cybersecurity agency is building a MongoDB database to track and analyze digital threats and cyber attacks worldwide. The database must handle diverse data, including timestamps, severity levels, attacker details, geolocation, and evidence files. Various MongoDB data types are used to store the following information efficiently:

- Stores attacker names and threat descriptions.
- Records severity scores and financial losses.
- Indicates whether the threat is resolved.
- Logs when an attack was detected.
- Stores IP addresses used in an attack.
- Uniquely identifies each attack record.
- Stores malware samples as encoded data.

Your job is to ensure the database efficiently handles various data types. Answer the following:

Q5(a): Insert a cyber attack record into a collection named "cyber_threats", ensuring the use of at least five different MongoDB data types. **(2 Marks)**

Q5(b): Query all cyber attacks that occurred in the last 30 days by using an appropriate MongoDB operator for the Date type. **(1 Mark)**

Q5(c): Remove a specific IP address ("192.168.1.10") from all attack records where it appears in an Array field. **(1 Mark)**